Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Kooistra Dairy

Physical address of dairy:

5837 Hultberg RD	Turlock	Stanislaus	95380
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 05/01/1980

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X045-X013-X026-XXXX X057-X017-X005-XXXX

B. OPERATORS

Kooistra, Cynthia Lea

Operator name: Kooistra, Cynthia Lea

Telephone no.: (209) 632-2311 Landline (209) 678-3112 Cellular

5837 Hultberg RD	Turlock	CA	95380
Mailing Address Number and Street	City	State	Zip Code

Kooistra, Sam Sikke

Operator name: Kooistra, Sam Sikke

Telephone no.: (209) 634-2311 Landline (209) 678-3114 Cellular

5837 Hultberg RD	Turlock	CA	95380
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Kooistra, Cynthia Lea

Legal owner name: Kooistra, Cynthia Lea

Telephone no.: (209) 632-2311 Landline (209) 678-3112 Cellular

5837 Hultberg RD	Turlock	CA	95380
Mailing Address Number and Street	City	State	Zip Code

Kooistra, Sam Sikke

Legal owner name:	Kooistra, Sam Sikke	Telephone no.:	(209) 634-2311	(209) 678-3114
			Landline	Cellular
5837 Hultberg RD	Turlock		CA	95380
Mailing Address Number and Street	City		State	Zip Code

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	35	155	155	145	0
Number under roof	375	0	0	0	0	0
Maximum number	375	35	155	155	145	0
Average number	375	35	155	155	145	0
Avg live weight (lbs)	1,250	1,300	800	600		

Predominant milk cow breed: Jersey

Average milk production: 72 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 13,636.05 *tons per reporting period*

Total nitrogen from manure: 170,134.38 *lbs per reporting period*

After ammonia losses (30% loss applied): 119,094.07 *lbs per reporting period*

Total phosphorus from manure: 27,849.00 *lbs per reporting period*

Total potassium from manure: 72,308.71 *lbs per reporting period*

Total salt from manure: 184,617.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 7,725,000 *gallons*

Total nitrogen generated: 75,326.25 *lbs*

Total phosphorus generated: 13,105.20 *lbs*

Total potassium generated: 91,842.98 *lbs*

Total salt generated: 488,015.60 *lbs*

	5,350,000	*gallons applied*
+	2,375,000	*gallons exported*
-	0	*gallons imported*
=	7,725,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Domestic Well - HFR-1	Ground water
Domestic Well - HFR-2	Ground water
Domestic Well - Home-1	Ground water
Domestic Well - Home-2	Ground water
Domestic Well - Wente	Ground water

Source Description	Type
Irrigation Water - Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/12/2023	Corral solids	1,400.00 *ton*	Dry-weight	20.4		17,000.00	10,100.00	11,600.00		0.00

Date	Material type	Quantity	Kjeldahl-N (mg/L)	Ammonium-N (mg/L)	Ammonia-N (mg/L)	Nitrate-N (mg/L)	P (mg/L)	K (mg/L)	EC (µmhos/cm)	TDS (mg/L)
07/27/2023	Process wastewater	775,000.00 *gal*	1,670.00	616.00	0.00	1.90	249.00	1,210.00		4,800
08/07/2023	Process wastewater	1,600,000.00 *gal*	1,670.00	616.00	0.00	1.90	249.00	1,210.00		4,800

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	37,889.60	22,510.88	25,854.08	0.00
Process wastewater	33,136.01	4,935.02	23,981.44	95,133.00
Total exports for all materials	71,025.61	27,445.90	49,835.52	95,133.00

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Fernandes	17	17	3	process wastewater	X045-X120-X013-XXXX
Kooistra	36	36	3	process wastewater	X045-X012-X011-XXXX X057-X018-X017-XXXX
Kooistra Dairy	10	10	0	none	X057-X017-X005-XXXX
Kooistra Heifer	10	10	3	process wastewater	X045-X013-X026-XXXX
Strickler	18	18	0	none	X044-X040-X003-XXXX
Totals for areas that were used for application	63	63	9		
Totals for areas that were not used for application	28	28	0		
Land application area totals	91	91	9		

B. CROPS AND HARVESTS

Fernandes

Field name: Fernandes

11/19/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 17 Plant date: 11/19/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	176.00 *ton*	Dry-weight		69.4	24,800.00	5,200.00	38,100.00		12.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	10.35	157.13	32.95	241.40	772.99

Fernandes

05/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 17 Plant date: 05/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/23/2023	540.20 *ton*	Dry-weight		71.3	12,600.00	2,800.00	14,800.00		5.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	31.78	229.82	51.07	269.95	930.22

09/09/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 17 Plant date: 09/09/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/24/2023	120.00 *ton*	Dry-weight		68.5	26,700.00	5,000.00	50,600.00		15.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	7.06	118.74	22.24	225.02	680.40

Kooistra

Field name: Kooistra

11/19/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 36 Plant date: 11/19/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	398.60 *ton*	Dry-weight		68.0	22,200.00	5,800.00	44,200.00		13.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.07	157.31	41.10	313.21	977.90

Kooistra

05/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 36 Plant date: 05/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/23/2023	1,007.10 *ton*	Dry-weight		70.6	14,600.00	2,800.00	18,500.00		6.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.98	240.16	46.06	304.31	1,019.86

09/09/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 36 Plant date: 09/09/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/24/2023	288.40 *ton*	Dry-weight		67.2	20,900.00	3,700.00	39,400.00		14.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	8.01	109.84	19.44	207.06	756.76

Kooistra Heifer

Field name: Kooistra Heifer

11/19/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 10 Plant date: 11/19/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	113.00 *ton*	Dry-weight		72.1	23,300.00	5,000.00	46,900.00		11.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.30	146.92	31.53	295.72	693.59

Kooistra Heifer

05/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 10 Plant date: 05/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
08/23/2023	298.30 *ton*	Dry-weight		77.0	16,500.00	3,300.00	21,700.00		7.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.83	226.41	45.28	297.76	1,015.41

09/09/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 10 Plant date: 09/09/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/24/2023	71.00 *ton*	Dry-weight		65.6	27,700.00	4,200.00	43,900.00		15.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	7.10	135.31	20.52	214.44	747.37

NUTRIENT BUDGET

A. LAND APPLICATIONS

Fernandes - 11/19/2022: Oats, silage-soft dough

Field name: Fernandes

Crop: Oats, silage-soft dough — Plant date: 11/19/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	113.65	22.01	182.80	1,058.34	245,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.91	2,525,625.00 *gal*
Application event totals		113.65	22.01	182.80	1,099.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	97.42	18.86	156.69	907.15	210,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.46	2,559,300.00 *gal*
Application event totals		97.42	18.86	156.69	948.61	

Fernandes - 05/17/2023: Corn, silage

Field name: Fernandes

Crop: Corn, silage — Plant date: 05/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.09	2,660,325.00 *gal*
Application event totals		0.00	0.00	0.00	43.09	

Fernandes - 05/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.55	2,626,650.00 *gal*
Application event totals		0.00	0.00	0.00	42.55	

06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	109.01	21.11	175.34	1,015.14	235,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.73	2,761,350.00 *gal*
Application event totals		109.01	21.11	175.34	1,059.88	

06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	109.01	21.11	175.34	1,015.14	235,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.28	2,795,025.00 *gal*
Application event totals		109.01	21.11	175.34	1,060.42	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.91	2,525,625.00 *gal*
Application event totals		0.00	0.00	0.00	40.91	

07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	97.42	18.86	156.69	907.15	210,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.37	2,491,950.00 *gal*
Application event totals		97.42	18.86	156.69	947.52	

Fernandes - 05/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.91	2,525,625.00 *gal*
Application event totals		0.00	0.00	0.00	40.91	

Fernandes - 09/09/2023: Sudangrass, silage

Field name: Fernandes

Crop: Sudangrass, silage

Plant date: 09/09/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	74.22	14.37	119.38	691.16	160,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.09	2,660,325.00 *gal*
Application event totals		74.22	14.37	119.38	734.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	69.58	13.47	111.92	647.96	150,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.55	2,626,650.00 *gal*
Application event totals		69.58	13.47	111.92	690.51	

Kooistra - 11/19/2022: Oats, silage-soft dough

Field name: Kooistra

Crop: Oats, silage-soft dough

Plant date: 11/19/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Kooistra - 11/19/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	109.53	21.21	176.17	1,019.94	500,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	38.12	4,983,900.00 *gal*
Application event totals		109.53	21.21	176.17	1,058.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	106.24	20.57	170.89	989.35	485,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.70	5,320,650.00 *gal*
Application event totals		106.24	20.57	170.89	1,030.05	

Kooistra - 05/17/2023: Corn, silage

Field name: Kooistra

Crop: Corn, silage

Plant date: 05/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	39.41	5,152,275.00 *gal*
Application event totals		0.00	0.00	0.00	39.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	38.38	5,017,575.00 *gal*
Application event totals		0.00	0.00	0.00	38.38	

Kooistra - 05/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	112.81	21.85	181.46	1,050.54	515,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	37.87	4,950,225.00 *gal*
Application event totals		112.81	21.85	181.46	1,088.41	

06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	108.43	21.00	174.41	1,009.75	495,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	37.87	4,950,225.00 *gal*
Application event totals		108.43	21.00	174.41	1,047.61	

07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	108.43	21.00	174.41	1,009.75	495,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	38.64	5,051,250.00 *gal*
Application event totals		108.43	21.00	174.41	1,048.38	

07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	38.38	5,017,575.00 *gal*
Application event totals		0.00	0.00	0.00	38.38	

08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.96	5,354,325.00 *gal*
Application event totals		0.00	0.00	0.00	40.96	

Kooistra - 09/09/2023: Sudangrass, silage

Field name: Kooistra

Crop: Sudangrass, silage Plant date: 09/09/2023

Kooistra - 09/09/2023: Sudangrass, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	64.62	12.51	103.94	601.77	295,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.50	5,556,375.00 *gal*
Application event totals		64.62	12.51	103.94	644.27	

10/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	62.43	12.09	100.42	581.37	285,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.22	5,388,000.00 *gal*
Application event totals		62.43	12.09	100.42	622.58	

Kooistra Heifer - 11/19/2022: Oats, silage-soft dough

Field name: Kooistra Heifer

Crop: Oats, silage-soft dough

Plant date: 11/19/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	102.52	19.85	164.90	954.67	130,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.30	1,717,425.00 *gal*
Application event totals		102.52	19.85	164.90	1,001.96	

03/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	98.58	19.09	158.56	917.95	125,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	51.00	1,852,125.00 *gal*
Application event totals		98.58	19.09	158.56	968.95	

Kooistra Heifer - 05/17/2023: Corn, silage

Kooistra Heifer - 05/17/2023: Corn, silage

Field name: Kooistra Heifer

Crop: Corn, silage

Plant date: 05/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.37	1,683,750.00 *gal*
Application event totals		0.00	0.00	0.00	46.37	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	39.88	1,448,025.00 *gal*
Application event totals		0.00	0.00	0.00	39.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	118.29	22.91	190.27	1,101.54	150,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.73	1,515,375.00 *gal*
Application event totals		118.29	22.91	190.27	1,143.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	98.58	19.09	158.56	917.95	125,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.30	1,717,425.00 *gal*
Application event totals		98.58	19.09	158.56	965.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	51.00	1,852,125.00 *gal*
Application event totals		0.00	0.00	0.00	51.00	

Kooistra Heifer - 05/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	82.80	16.03	133.19	771.08	105,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	36.17	1,313,325.00 *gal*
Application event totals		82.80	16.03	133.19	807.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.44	1,650,075.00 *gal*
Application event totals		0.00	0.00	0.00	45.44	

Kooistra Heifer - 09/09/2023: Sudangrass, silage

Field name: Kooistra Heifer

Crop: Sudangrass, silage

Plant date: 09/09/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	90.69	17.56	145.87	844.51	115,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.30	1,717,425.00 *gal*
Application event totals		90.69	17.56	145.87	891.81	

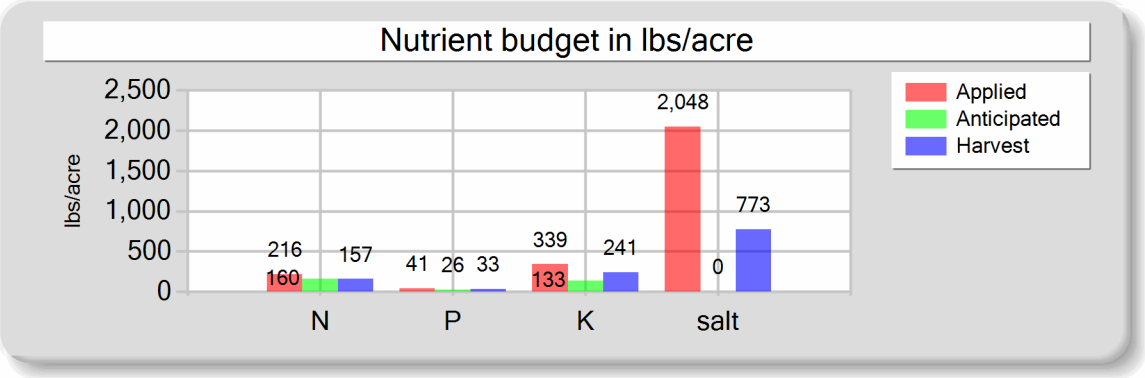
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	67.03	12.98	107.82	624.21	85,000.00 *gal*
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.44	1,650,075.00 *gal*
Application event totals		67.03	12.98	107.82	669.65	

B. NUTRIENT BUDGET

Fernandes - 11/19/2022: Oats, silage-soft dough

Field name: Fernandes Crop: Oats, silage-soft dough Plant date: 11/19/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	211.07	40.87	339.49	1,965.49
Fresh water	0.00	0.00	0.00	82.37
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	215.73	40.87	339.49	2,047.86
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	157.13	32.95	241.40	772.99
Nutrient balance	58.60	7.93	98.09	1,274.87
Applied to removed ratio	1.37	1.24	1.41	2.65

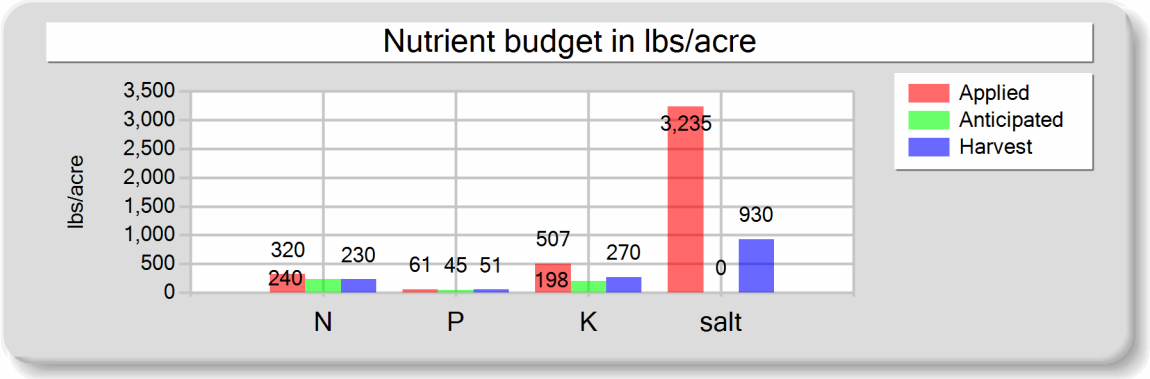
Fresh water applied	
5,084,925.00	*gallons*
187.26	*acre-inches*
11.02	*inches/acre*

Process wastewater applied	
455,000.00	*gallons*
16.76	*acre-inches*
0.99	*inches/acre*

Total harvests for the crop	
1	*harvests*

Fernandes - 05/17/2023: Corn, silage

Field name: Fernandes Crop: Corn, silage Plant date: 05/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	315.44	61.09	507.38	2,937.44
Fresh water	0.00	0.00	0.00	297.85
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	320.11	61.09	507.38	3,235.29
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	229.82	51.07	269.95	930.22
Nutrient balance	90.29	10.01	237.43	2,305.06
Applied to removed ratio	1.39	1.20	1.88	3.48

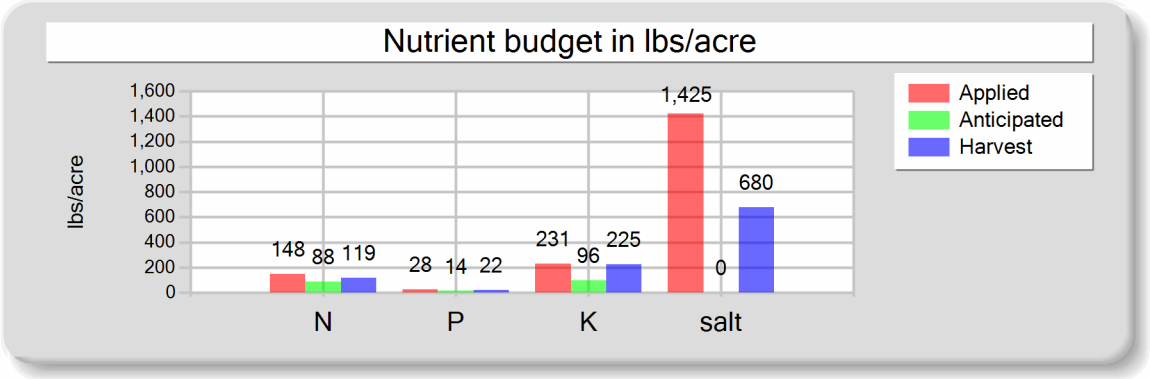
Fresh water applied	
18,386,550.00	*gallons*
677.11	*acre-inches*
39.83	*inches/acre*

Process wastewater applied	
680,000.00	*gallons*
25.04	*acre-inches*
1.47	*inches/acre*

Total harvests for the crop	
1	*harvests*

Fernandes - 09/09/2023: Sudangrass, silage

Field name: Fernandes | Crop: Sudangrass, silage | Plant date: 09/09/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	143.80	27.85	231.30	1,339.13
Fresh water	0.00	0.00	0.00	85.64
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	148.47	27.85	231.30	1,424.77
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	118.74	22.24	225.02	680.40
Nutrient balance	29.73	5.61	6.28	744.37
Applied to removed ratio	1.25	1.25	1.03	2.09

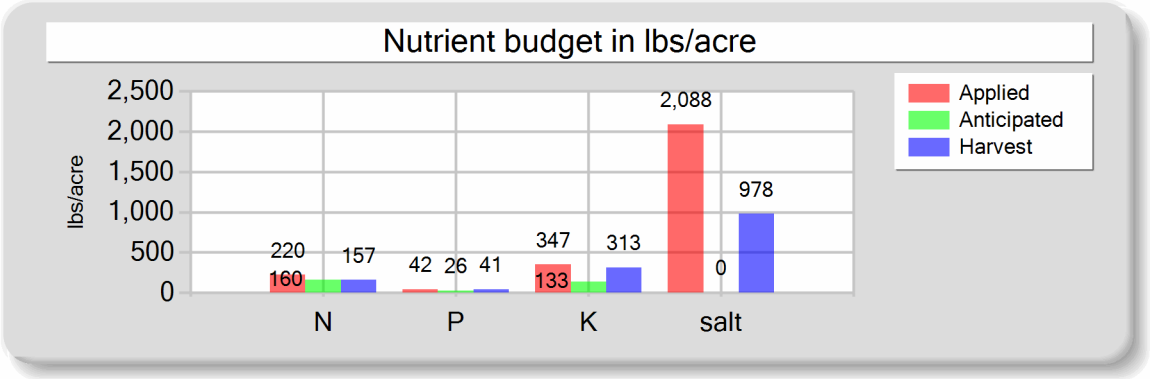
Fresh water applied	
5,286,975.00	*gallons*
194.70	*acre-inches*
11.45	*inches/acre*

Process wastewater applied	
310,000.00	*gallons*
11.42	*acre-inches*
0.67	*inches/acre*

Total harvests for the crop	
1	*harvests*

Kooistra - 11/19/2022: Oats, silage-soft dough

Field name: Kooistra | Crop: Oats, silage-soft dough | Plant date: 11/19/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	215.77	41.78	347.06	2,009.29
Fresh water	0.00	0.00	0.00	78.83
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	220.44	41.78	347.06	2,088.12
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	157.31	41.10	313.21	977.90
Nutrient balance	63.12	0.68	33.85	1,110.22
Applied to removed ratio	1.40	1.02	1.11	2.14

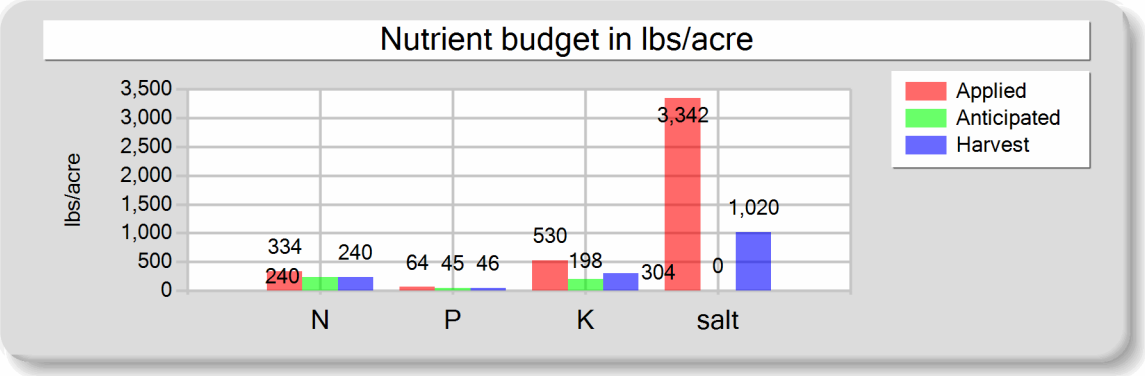
Fresh water applied	
10,304,550.00	*gallons*
379.48	*acre-inches*
10.54	*inches/acre*

Process wastewater applied	
985,000.00	*gallons*
36.27	*acre-inches*
1.01	*inches/acre*

Total harvests for the crop	
1	*harvests*

Kooistra - 05/17/2023: Corn, silage

Field name: Kooistra Crop: Corn, silage Plant date: 05/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	329.68	63.84	530.28	3,070.03
Fresh water	0.00	0.00	0.00	271.51
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	334.35	63.84	530.28	3,341.54
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	240.16	46.06	304.31	1,019.86
Nutrient balance	94.19	17.78	225.97	2,321.69
Applied to removed ratio	1.39	1.39	1.74	3.28

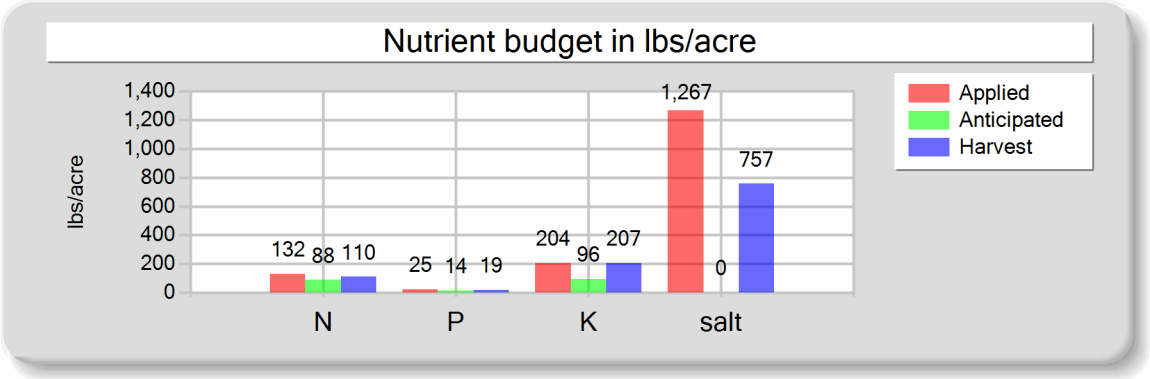
Fresh water applied	
35,493,450.00	*gallons*
1,307.10	*acre-inches*
36.31	*inches/acre*

Process wastewater applied	
1,505,000.00	*gallons*
55.42	*acre-inches*
1.54	*inches/acre*

Total harvests for the crop	
1	*harvests*

Kooistra - 09/09/2023: Sudangrass, silage

Field name: Kooistra | Crop: Sudangrass, silage | Plant date: 09/09/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	127.05	24.60	204.36	1,183.14
Fresh water	0.00	0.00	0.00	83.72
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	131.72	24.60	204.36	1,266.86
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	109.84	19.44	207.06	756.76
Nutrient balance	21.88	5.16	-2.70	510.09
Applied to removed ratio	1.20	1.27	0.99	1.67

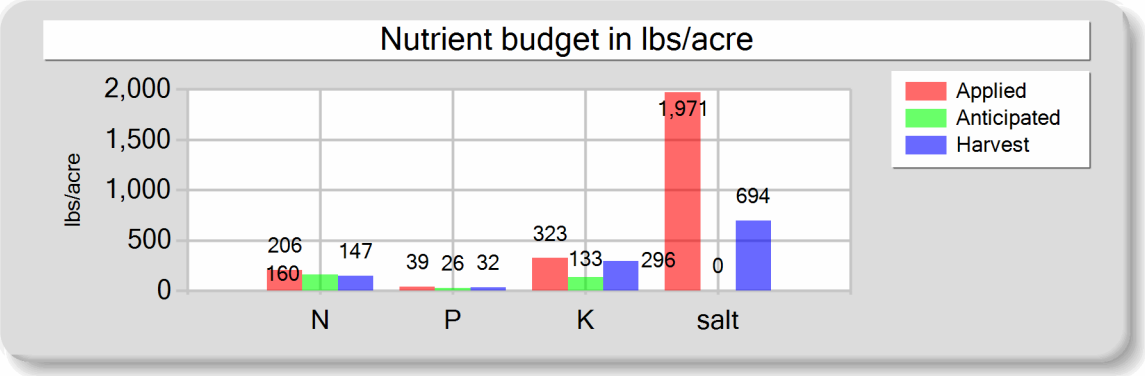
Fresh water applied	
10,944,375.00	*gallons*
403.04	*acre-inches*
11.20	*inches/acre*

Process wastewater applied	
580,000.00	*gallons*
21.36	*acre-inches*
0.59	*inches/acre*

Total harvests for the crop	
1	*harvests*

Kooistra Heifer - 11/19/2022: Oats, silage-soft dough

Field name: Kooistra Heifer Crop: Oats, silage-soft dough Plant date: 11/19/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	201.09	38.94	323.45	1,872.62
Fresh water	0.00	0.00	0.00	98.30
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	205.76	38.94	323.45	1,970.92
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	146.92	31.53	295.72	693.59
Nutrient balance	58.84	7.41	27.73	1,277.32
Applied to removed ratio	1.40	1.24	1.09	2.84

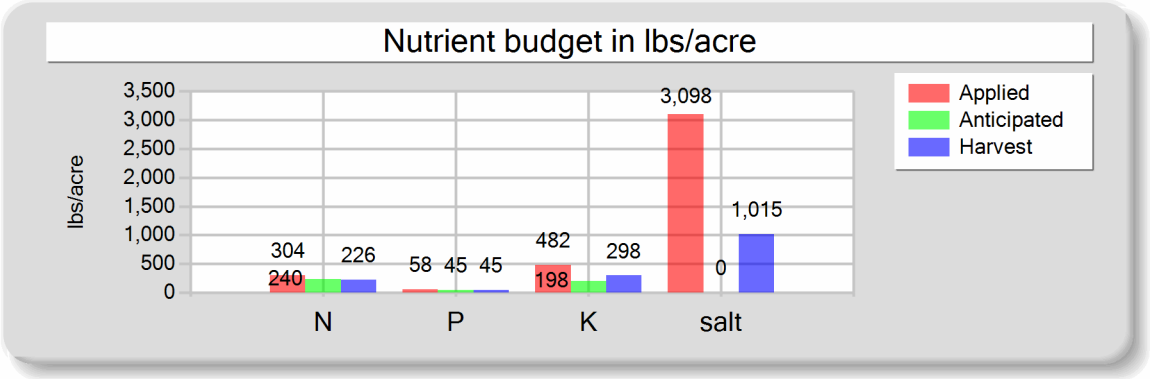
Fresh water applied	
3,569,550.00	*gallons*
131.45	*acre-inches*
13.15	*inches/acre*

Process wastewater applied	
255,000.00	*gallons*
9.39	*acre-inches*
0.94	*inches/acre*

Total harvests for the crop	
1	*harvests*

Kooistra Heifer - 05/17/2023: Corn, silage

Field name: Kooistra Heifer Crop: Corn, silage Plant date: 05/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	299.67	58.03	482.01	2,790.57
Fresh water	0.00	0.00	0.00	307.88
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	304.34	58.03	482.01	3,098.45
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	226.41	45.28	297.76	1,015.41
Nutrient balance	77.93	12.75	184.24	2,083.04
Applied to removed ratio	1.34	1.28	1.62	3.05

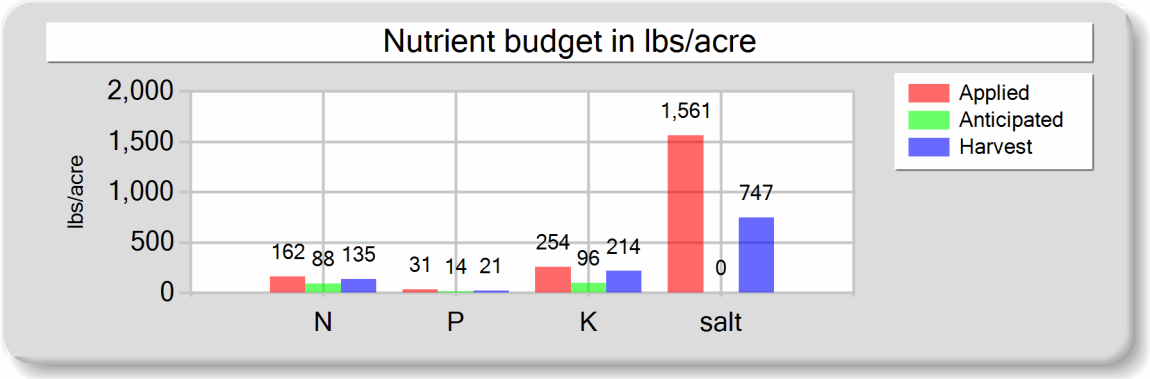
Fresh water applied	
11,180,100.00	*gallons*
411.73	*acre-inches*
41.17	*inches/acre*

Process wastewater applied	
380,000.00	*gallons*
13.99	*acre-inches*
1.40	*inches/acre*

Total harvests for the crop	
1	*harvests*

Kooistra Heifer - 09/09/2023: Sudangrass, silage

Field name: Kooistra Heifer Crop: Sudangrass, silage Plant date: 09/09/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	157.72	30.54	253.69	1,468.72
Fresh water	0.00	0.00	0.00	92.74
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	162.39	30.54	253.69	1,561.46
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	135.31	20.52	214.44	747.37
Nutrient balance	27.08	10.03	39.25	814.08
Applied to removed ratio	1.20	1.49	1.18	2.09

Fresh water applied	
3,367,500.00	*gallons*
124.01	*acre-inches*
12.40	*inches/acre*

Process wastewater applied	
200,000.00	*gallons*
7.37	*acre-inches*
0.74	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Solid Manure

Sample and source description: Solid Manure

Sample date: 05/15/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 20.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,000.00	10,100.00	11,600.00							
DL	100.00	100.00	100.00							

Solid Manure

Sample and source description: Solid Manure

Sample date: 11/01/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 52.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,500.00	9,500.00	24,300.00							
DL	100.00	100.00	100.00							

B. PROCESS WASTEWATER ANALYSES

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 02/20/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,670.00	616.00	0.00	1.90	249.00	1,210.00								13,900.00	4,800
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Waste water

Sample and source description: Waste water

Sample date: 06/14/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,150.00	548.00	0.00	1.10	158.00	2,010.00								17,200.00	7,600
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 09/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,060.00	332.00	0.00	0.80	235.00	1,790.00								13,400.00	9,750
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 11/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.30

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	945.00	229.00	0.00	0.00	183.00	1,520.00								10,800.00	8,800
DL	1.00	0.50	1.00	1.00	0.10	0.50								10.00	10

C. FRESH WATER ANALYSES

Domestic Well - HFR-1

Domestic Well - HFR-1

HFR-1

Sample description: HFR-1

Sample date: 11/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	32.50	0.00	32.50								1,080.00	0
DL	1.00	1.00	0.10								10.00	1

Domestic Well - HFR-2

HFR-2

Sample description: HFR-2

Sample date: 11/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	43.40	0.00	43.40								1,200.00	0
DL	1.00	1.00	0.10								10.00	1

Domestic Well - Home-1

Home-1

Sample description: Home-1

Sample date: 11/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								1,840.00	0
DL	1.00	1.00	1.00								10.00	1

Domestic Well - Home-2

Domestic Well - Home-2

Home-2

Sample description: Home-2

Sample date: 11/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								2,040.00	0
DL	1.00	1.00	1.00								10.00	1

Domestic Well - Wente

Wente

Sample description: Wente

Sample date: 11/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	20.30	0.00	20.30								984.00	0
DL	1.00	1.00	0.10								10.00	1

Irrigation Water - Canal

Canal

Sample description: Canal

Sample date: 07/17/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								33.20	33
DL	1.00	1.00	1.00								10.00	10

D. SOIL ANALYSES

Fernandes

Fernandes

Sample and source description: Fernandes

Sample date: 11/29/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			61.00				
DL			2.00				

Kooistra

Kooistra

Sample and source description: Kooistra

Sample date: 11/29/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			324.00				
DL			2.00				

Kooistra Heifer

Kooistra-Heifer

Sample and source description: Kooistra-Heifer

Sample date: 11/29/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			76.00				
DL			2.00				

E. PLANT TISSUE ANALYSES

Fernandes - 11/19/2022: Oats, silage-soft dough

Fernandes - 11/19/2022: Oats, silage-soft dough

Fernandes

Sample and source description: Fernandes

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,800.00	5,200.00	38,100.00		12.20
DL	100.00	100.00	100.00		0.01

Fernandes - 05/17/2023: Corn, silage

Fernandes

Sample and source description: Fernandes

Sample date: 08/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,600.00	2,800.00	14,800.00		5.10
DL	100.00	100.00	100.00		0.01

Fernandes - 09/09/2023: Sudangrass, silage

Fernandes

Sample and source description: Fernandes

Sample date: 10/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,700.00	5,000.00	50,600.00		15.30
DL	100.00	100.00	100.00		0.01

Kooistra - 11/19/2022: Oats, silage-soft dough

Kooistra

Sample and source description: Kooistra

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,200.00	5,800.00	44,200.00		13.80
DL	100.00	100.00	100.00		0.01

Kooistra - 05/17/2023: Corn, silage

Kooistra

Sample and source description: Kooistra

Sample date: 08/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,600.00	2,800.00	18,500.00		6.20
DL	100.00	100.00	100.00		0.01

Kooistra - 09/09/2023: Sudangrass, silage

Kooistra

Sample and source description: Kooistra

Sample date: 10/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,900.00	3,700.00	39,400.00		14.40
DL	100.00	100.00	100.00		0.01

Kooistra Heifer - 11/19/2022: Oats, silage-soft dough

Kooistra Heifer

Sample and source description: Kooistra Heifer

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,300.00	5,000.00	46,900.00		11.00
DL	100.00	100.00	100.00		0.01

Kooistra Heifer - 05/17/2023: Corn, silage

Kooistra Heifer

Sample and source description: Kooistra Heifer

Sample date: 08/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 77.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,500.00	3,300.00	21,700.00		7.40
DL	100.00	100.00	100.00		0.01

Kooistra Heifer - 09/09/2023: Sudangrass, silage

Kooistra Heifer

Sample and source description: Kooistra Heifer

Sample date: 10/24/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,700.00	4,200.00	43,900.00		15.30
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

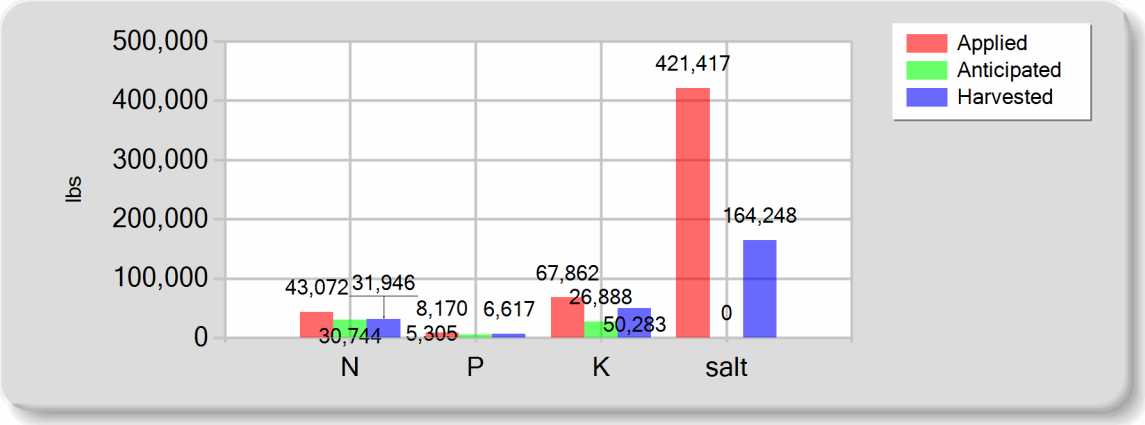
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

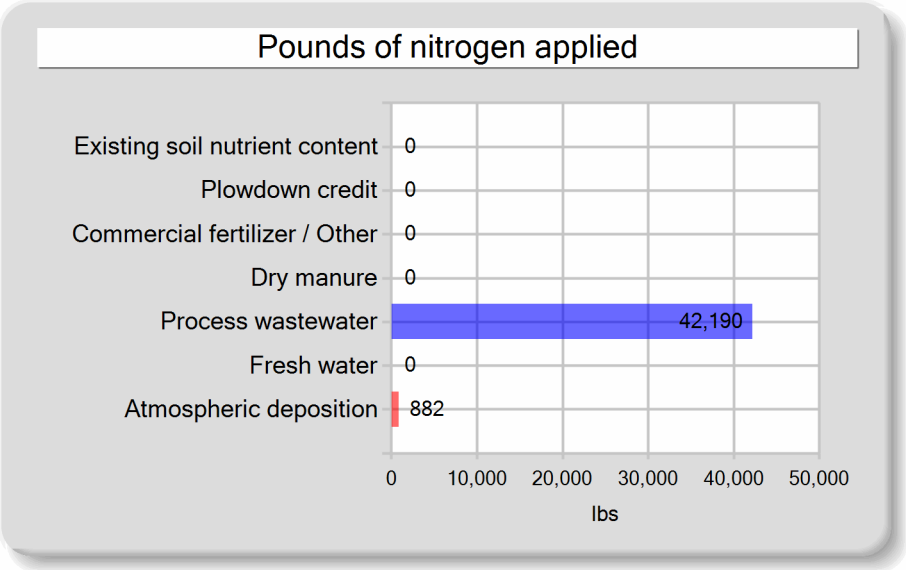
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

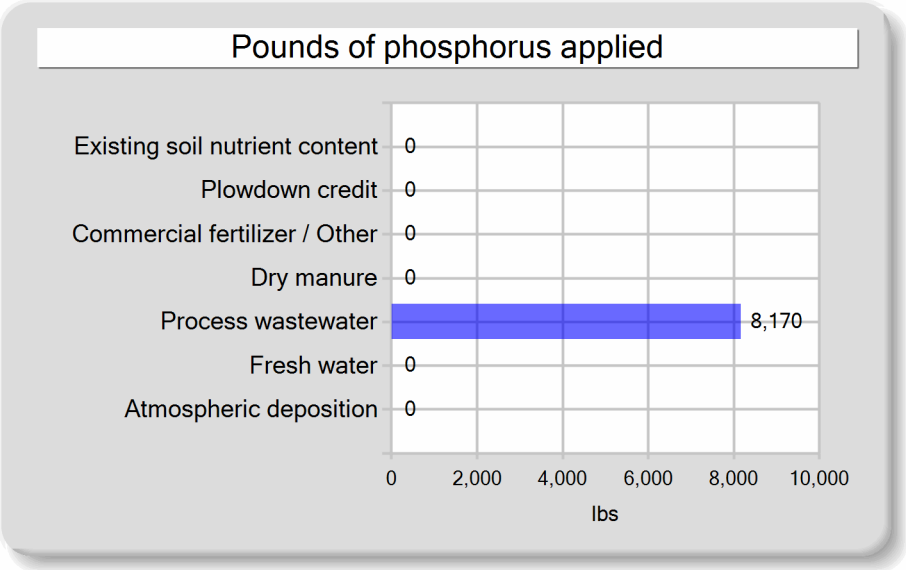
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	42,190.23	8,170.17	67,861.54	392,882.60
Fresh water	0.00	0.00	0.00	28,534.84
Atmospheric deposition	882.00	0.00	0.00	0.00
Total nutrients applied	43,072.23	8,170.17	67,861.54	421,417.44
Anticipated crop nutrient removal	30,744.00	5,304.60	26,888.40	0.00
Actual crop nutrient removal	31,946.21	6,617.26	50,282.52	164,247.90
Nutrient balance	11,126.03	1,552.91	17,579.02	257,169.53
Applied to removed ratio	1.35	1.23	1.35	2.57

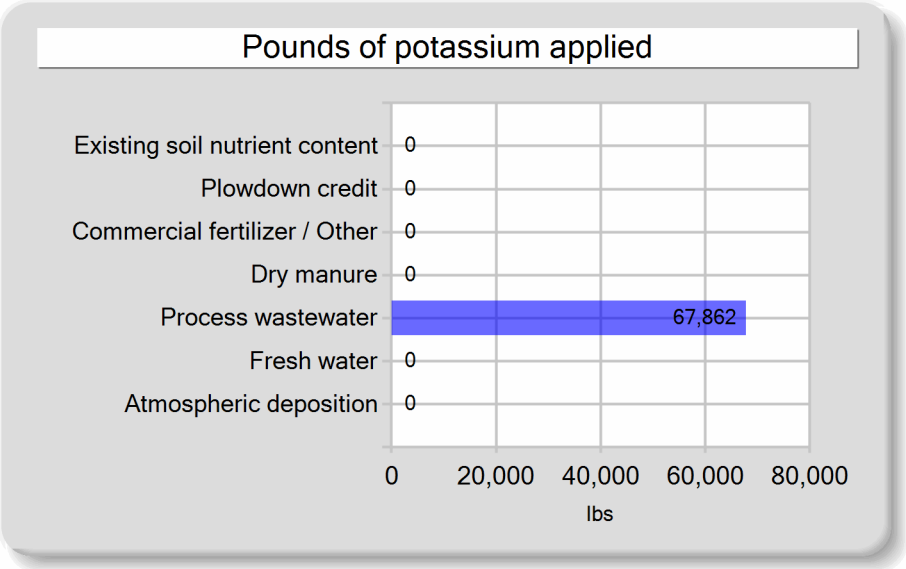
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

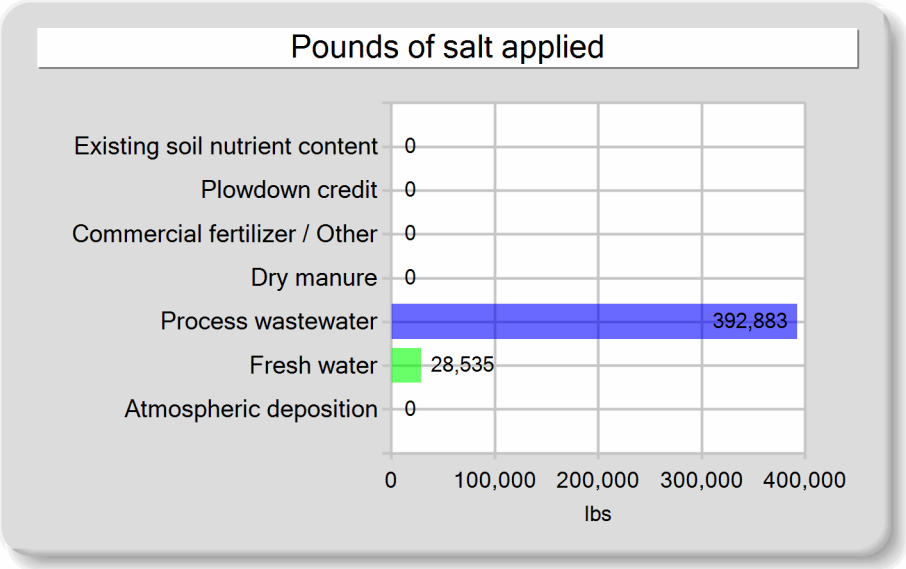


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Sam Kooistra (Jun 21, 2024 11:03 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Sam Sikke Kooistra	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
06/21/2024	
DATE	DATE

ATTACHMENT D

Manure Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: **Sam Kooistra**

Name of Dairy Facility: **Kooistra Dairy**

Facility Address: **5837 Hultberg Road, Turlock, CA 95380**

Contact Person Name and Phone Number: **Sam Kooistra (209) 678-3114**

Manure Hauler Information:

Name of Hauling Company/Person: **Five Star Custom Spreading**

Address of Hauling Company /Person: **P.O. Box 488, Gustine, CA 95322**

Contact Person and Phone Number: **Eric Rocha (209) 678-2917**

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) **Broker** (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Number and Street	City	Zip Code	Phone Number
Five Star Custom Spreading	P.O. Box 488	Gustine	95322	(209) 678-2917

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Number and Street	City	Zip Code	Assessor's Parcel Number
750 East Ave	Gustine	95322	

Dates Hauled: **10/12/23**

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: **1,400** Tons

Manure Solids Content (if amount reported in tons): **79.6%**

Method used to determine amount of manure: _____

Number of Truckloads x Average Truck Weight

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: **N/A** Gallons

Method used to determine volume of process wastewater: **N/A**

Written Agreement:

****NOT NECESSARY FOR SOLID MANURE MANIFEST****

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Sam Kooistra (Jun 21, 2024 11:03 PDT) Date: 06/21/2024

Hauler's Signature: _____ Date: 10/12/2023

ATTACHMENT D

Manure Tracking Manifest
For
Existing Milk Cow Dairies

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1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: **Sam Kooistra**

Name of Dairy Facility: **Kooistra Dairy**

Facility Address: **5837 Hultberg Road, Turlock, CA 95380**

Contact Person Name and Phone Number: **Sam Kooistra (209) 678-3114**

Manure Hauler Information:

Name of Hauling Company/Person: **Kooistra Dairy**

Address of Hauling Company /Person: **5837 Hultberg Road, Turlock, CA 95380**

Contact Person and Phone Number: **Sam Kooistra (209) 678-3114**

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) **Farmer** (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Number and Street	City	Zip Code	Phone Number
Elmiro Moitoso	**5701 Hultberg Road**	**Hilmar**	**95324**	**(209) 632-1873**

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Number and Street	City	Zip Code	Assessor's Parcel Number
	Stanislaus County		**X057-X017-X004-XXXX**

Dates Hauled: **7/1/23, 7/14/23, & 7/27/23**

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____Tons

Manure Solids Content (if amount reported in tons): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: **775,000** Gallons

Method used to determine volume of process wastewater: _____

Total Run Time x Pump Output GPM

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Sam Kooistra (Jun 21, 2024 11:03 PDT) Date: 06/21/2024

Hauler's Signature: Sam Kooistra (Jun 21, 2024 11:03 PDT) Date: 06/21/2024

ATTACHMENT D

Manure Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

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2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: **Sam Kooistra**

Name of Dairy Facility: **Kooistra Dairy**

Facility Address: **5837 Hultberg Road, Turlock, CA 95380**

Contact Person Name and Phone Number: **Sam Kooistra (209) 678-3114**

Manure Hauler Information:

Name of Hauling Company/Person: **Kooistra Dairy**

Address of Hauling Company /Person: **5837 Hultberg Road, Turlock, CA 95380**

Contact Person and Phone Number: **Sam Kooistra (209) 678-3114**

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) **Farmer** (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Name	Number and Street	City	Zip Code	Phone Number
Fernando Toste	**9865 Tegner Road**	**Hilmar**	**95324**	**(209) 495-1041**

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

Number and Street	City	Zip Code	Assessor's Parcel Number
	Merced County		**X045-X013-X027-XXXX**

Dates Hauled: **3/17/23, 7/9/23, 7/22/23, & 8/7/23**

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: _____Tons

Manure Solids Content (if amount reported in tons): _____

Method used to determine amount of manure: _____

_____

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: **1,600,000** Gallons

Method used to determine volume of process wastewater: _____

Total Run Time x Pump Output GPM

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

X Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Sam Kooistra (Jun 21, 2024 11:03 PDT) Date: 06/21/2024

Hauler's Signature: Sam Kooistra (Jun 21, 2024 11:03 PDT) Date: 06/21/2024

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Sam Kooistra Dairy

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:31

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23K0678-01	HFR-1	Ag Water	T. Madruga		11/13/2023 15:00
23K0678-02	HFR-2	Ag Water	T. Madruga		11/13/2023 15:15
23K0678-03	Home-1	Ag Water	T. Madruga		11/13/2023 15:30
23K0678-04	Home-2	Ag Water	T. Madruga		11/13/2023 15:45
23K0678-05	Wente	Ag Water	T. Madruga		11/13/2023 16:00

Default Cooler Temperature on Receipt °C: 2.6
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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Ranch: Sam Kooistra Dairy

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:31

Sample Results

Sample: HFR-1
23K0678-01 (Water)

Sampled: 11/13/2023 15:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.08**	mmhos/cm	0.01	1		11/17/23 11:18	SM 2510 B		BEK0478
Electrical Conductivity umhos	**1080**	umhos/cm	10.0	1		11/17/23 11:18	SM 2510 B		BEK0478
Ammonia (as N)	ND	mg/L	0.00	1		11/13/23 15:00	Field		BEK0425
Nitrate Nitrogen as NO3N	**32.5**	mg/L	0.1	1	10	11/14/23 21:08	EPA 300.0		BEK0466
Temperature	**25.0**	units	0.0	1		11/17/23 11:18	SM 4500-H+	H	BEK0478
pH	**7.9**	units	1.0	1		11/17/23 11:18	SM 4500-H+	H	BEK0478

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Received: 11/14/2023 15:45
Reported: 11/21/2023 09:31

Sample Results
(Continued)

Sample: HFR-2
23K0678-02 (Water)

Sampled: 11/13/2023 15:15
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.20**	mmhos/cm	0.01	1		11/17/23 11:19	SM 2510 B		BEK0478
Electrical Conductivity umhos	**1200**	umhos/cm	10.0	1		11/17/23 11:19	SM 2510 B		BEK0478
Ammonia (as N)	ND	mg/L	0.00	1		11/13/23 15:15	Field		BEK0425
Nitrate Nitrogen as NO3N	**43.4**	mg/L	0.1	1	10	11/14/23 21:28	EPA 300.0		BEK0466
Temperature	**25.0**	units	0.0	1		11/17/23 11:19	SM 4500-H+	H	BEK0478
pH	**7.8**	units	1.0	1		11/17/23 11:19	SM 4500-H+	H	BEK0478

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Hilmar, CA 95324

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Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Sam Kooistra Dairy

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:31

Sample Results
(Continued)

Sample: Home-1
23K0678-03 (Water)

Sampled: 11/13/2023 15:30
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.84**	mmhos/cm	0.01	1		11/17/23 11:21	SM 2510 B		BEK0478
Electrical Conductivity umhos	**1840**	umhos/cm	10.0	1		11/17/23 11:21	SM 2510 B		BEK0478
Ammonia (as N)	ND	mg/L	0.00	1		11/13/23 15:30	Field		BEK0425
Nitrate Nitrogen as NO3N	ND	mg/L	0.1	1	10	11/14/23 21:47	EPA 300.0		BEK0466
Temperature	**25.0**	units	0.0	1		11/17/23 11:21	SM 4500-H+	H	BEK0478
pH	**7.8**	units	1.0	1		11/17/23 11:21	SM 4500-H+	H	BEK0478

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Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Sam Kooistra Dairy

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:31

Sample Results
(Continued)

Sample: Home-2
23K0678-04 (Water)

Sampled: 11/13/2023 15:45
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**2.04**	mmhos/cm	0.01	1		11/17/23 11:22	SM 2510 B		BEK0478
Electrical Conductivity umhos	**2040**	umhos/cm	10.0	1		11/17/23 11:22	SM 2510 B		BEK0478
Ammonia (as N)	ND	mg/L	0.00	1		11/13/23 15:45	Field		BEK0425
Nitrate Nitrogen as NO3N	ND	mg/L	0.1	1	10	11/14/23 22:07	EPA 300.0		BEK0466
Temperature	**25.0**	units	0.0	1		11/17/23 11:22	SM 4500-H+	H	BEK0478
pH	**7.8**	units	1.0	1		11/17/23 11:22	SM 4500-H+	H	BEK0478

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LABORATORY INC

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Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Sam Kooistra Dairy

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:31

Sample Results
(Continued)

Sample: Wente
23K0678-05 (Water)

Sampled: 11/13/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.98**	mmhos/cm	0.01	1		11/17/23 11:24	SM 2510 B		BEK0478
Electrical Conductivity umhos	**984**	umhos/cm	10.0	1		11/17/23 11:24	SM 2510 B		BEK0478
Ammonia (as N)	ND	mg/L	0.00	1		11/13/23 16:00	Field		BEK0425
Nitrate Nitrogen as NO3N	**20.3**	mg/L	0.1	1	10	11/14/23 22:27	EPA 300.0		BEK0466
Temperature	**25.0**	units	0.0	1		11/17/23 11:24	SM 4500-H+	H	BEK0478
pH	**7.9**	units	1.0	1		11/17/23 11:24	SM 4500-H+	H	BEK0478

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Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Sam Kooistra Dairy

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:31

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0466										
Blank (BEK0466-BLK1)					Prepared & Analyzed: 11/14/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0466-BLK2)					Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0466-BLK3)					Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEK0466-BS1)					Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		103	90-110		
LCS (BEK0466-BS2)					Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		104	90-110		
Duplicate (BEK0466-DUP1)		**Source: 23K0678-05**			Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	20.3		0.1	mg/L		20.3			0.133	10
Duplicate (BEK0466-DUP2)		**Source: 23K0691-01**			Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	31.0		0.1	mg/L		31.3			1.08	10
Matrix Spike (BEK0466-MS1)		**Source: 23K0678-05**			Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	25.8		0.1	mg/L	5.000	20.3	110	90-110		
Matrix Spike (BEK0466-MS2)		**Source: 23K0691-01**			Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	36.1		0.1	mg/L	5.000	31.3	96.8	90-110		
Reference (BEK0466-SRM1)					Prepared & Analyzed: 11/14/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEK0466-SRM2)					Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEK0466-SRM3)					Prepared: 11/14/2023 Analyzed: 11/15/2023					
Nitrate Nitrogen as NO3N	10.4			mg/L	10.00		104	90-110		

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Ranch: Sam Kooistra Dairy

Received: 11/14/2023 15:45
Reported: 11/21/2023 09:31

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0478										
Blank (BEK0478-BLK1)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	6.0		1.0	units						
Blank (BEK0478-BLK2)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.1		1.0	units						
Blank (BEK0478-BLK3)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.5		1.0	units						
Duplicate (BEK0478-DUP1)		**Source: 23K0682-02**			Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	2.44		0.01	mmhos/cm		2.44			0.0205	10
pH	7.8		1.0	units		7.8			0.128	10
Electrical Conductivity umhos	2440		10.0	umhos/cm		2440			0.0205	10
Duplicate (BEK0478-DUP2)		**Source: 23K0690-02**			Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	1.19		0.01	mmhos/cm		1.18			0.304	10
Electrical Conductivity umhos	1190		10.0	umhos/cm		1180			0.304	10
pH	7.8		1.0	units		7.8			0.128	10
Reference (BEK0478-SRM1)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	562			umhos/cm	538.0		105	90-110		
Reference (BEK0478-SRM2)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEK0478-SRM3)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	1060			umhos/cm	1000		106	90-110		
Electrical Conductivity umhos	1060			umhos/cm	1000		106	90-110		
Reference (BEK0478-SRM4)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	1060			umhos/cm	1000		106	90-110		
Electrical Conductivity umhos	1060			umhos/cm	1000		106	90-110		
Reference (BEK0478-SRM5)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity	1080			umhos/cm	1000		108	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0478 (Continued)										
Reference (BEK0478-SRM5)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
Electrical Conductivity umhos	1080			umhos/cm	1000		108	90-110		
Reference (BEK0478-SRM6)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEK0478-SRM7)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEK0478-SRM8)					Prepared: 11/15/2023	Analyzed: 11/17/2023				
pH	4.0			units	4.000		100	97.5-102.5		

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11/14/23 15:45 23K0678

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

JG

No. Samples 5 No. of Bottles: 5

Purchase Order No | Bill To: Acct # 19840 | Cons # 08

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Results Need By

Name: R & J Avila Inc
Address: 8119 Ellington Avenue
City: Hilmar **State:** CA **Zip:** 95324
Telephone: (209) 669-7919 **Fax:** (209) 634-4604
Cell/Email: 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com

COPY TO:

REQUESTED BY: Rui & John Avila
PROJECT Sam Kooistra Dairy
ID CROP: N/A

[X] Copy of Chain [X] QA/QC Documents

Sampled By: T. Madruga

Other:

Analysis and Bottles Required: (Please indicate Analysis)

(X) DWW1: EC, pH, NO_3-N (NH4-N Field Test)
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DCW1: EC, NO_3-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	HFR-1	11/13/2023	1500	2.6	ND
2	HFR-2	11/13/2023	1515	2.6	ND
3	Home-1	11/13/2023	1530	1.4	ND
4	Home-2	11/13/2023	1545	2.5	ND
5	Wente	11/13/2023	1600	3.3	ND
6					
7					
8					
9					
10					

IR Thermometer SN: 192603727
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	John Avila	R & J Avila, Inc.	11/13/2023 @ 1600	11/14/2023 @ 900
Second	JLP	Dellavalle Laboratory, Inc.	11/14/2023 @ 900	11-14-23 3:30
Third				
Fourth	[illegible]	DVJ	11/14 15:45	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through ***cal*** under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration,reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information: ***Shipping***
Sampling hrs $ In
Miles $ Out
Consulting

Amt Paid	Rec By	Check #	Date

Signature

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

11/14/23 15:45 23K0678

Shipping Information: Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

Shipping Container: Ice Chest ☒ None ☐ Box ☐

Refrigerant: Wet Ice: ☒ Blue Ice: ☐ None: ☐

Container(s) Received	1	2	3	4	5	6	7	8	9	10
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)										
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	1	1	1	1	1					
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										